GALIYAN PINDI DIYAN

AN INTERACTIVE 3D & VR CULTURAL EXPERIENCE

Galiyan Pindi Diyan is an interactive VR rickshaw journey game through Androon Rawalpindi, where players explore its architecture, culture and memories. One street, stop and story at a time.

Project Overview

The project transforms my original 3D animated film into an experience where people are not simply watching the city, they are exploring it. Users select a driver, choose a passenger and decide which route they want to experience. During the journey, they can observe historic architecture, stop at important locations and listen to stories connected to the area.

The aim is to preserve the identity of old Rawalpindi while making its history approachable, personal and engaging for a younger generation.

Project Type: Interaction Design, 3D Environment and VR Experience

Tools: Blender, Figma, Adobe After Effects, Adobe Illustrator, Adobe Photoshop and Unity.

Platform: Desktop prototype and VR experience

Target Audience: Young adults, students, tourists and people connected to Rawalpindi

THE BEGINNING

An Interactive 3D & VR Cultural Experience

This project began as my **University Thesis**: a 3D animated journey through Androon Rawalpindi.
The original animation followed a rickshaw moving through the old city on a winter night. It presented narrow streets, historic buildings, food stalls, warm lights, smoke, puddles, roadside shops and the everyday movement of the city.
The film successfully communicated the atmosphere of Rawalpindi, but the viewer could only follow a fixed camera and predetermined story.

This created a new question!

What if people could enter this world, sit inside the rickshaw and decide how they wanted to experience the city?

That question became the foundation of Galiyan Pindi Diyan.

The Problem

The viewer watches what the filmmaker chooses to show, **follows a fixed sequence and has little control over the experience.** Young audiences are increasingly familiar with interactive media, games and virtual environments. A standard documentary may provide information, but it does not always create the feeling of discovery that makes a place memorable.

DESIGN CHALLENGE

How might I transform a linear **3D animation** into an interactive cultural journey that allows people to explore Rawalpindi through choice, movement and conversation?

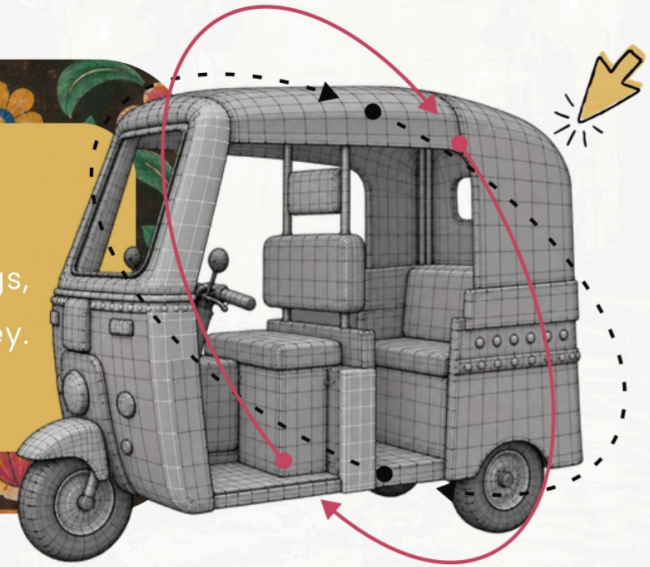
PROJECT AIM

The aim of Galiyan Pindi Diyan is to create an immersive cultural experience that allows users to discover Androon Rawalpindi from inside a moving rickshaw.

The experience combines:

- 3D environmental storytelling
- VR exploration
- Character selection
- Route selection
- Interactive conversations
- Architectural information
- Local memories and stories
- Pakistani visual culture
- Ambient sound and spatial audio

The **rickshaw** acts as both transportation and a storytelling space. It connects different streets, buildings, characters and memories within one continuous journey.



RESEARCH QUESTIONS

The research was guided by five main questions:

1. How familiar are people with the historic buildings and cultural locations of Androon Rawalpindi?
2. What prevents people from visiting or learning about these places?
3. Which parts of Rawalpindi architecture, food, markets, memories or stories interest people the most?
4. Would people prefer to explore Rawalpindi through a video, website, game or immersive VR journey?

RESEARCH PROCESS

I used a mixed research process that included both primary and secondary research.

PRIMARY RESEARCH

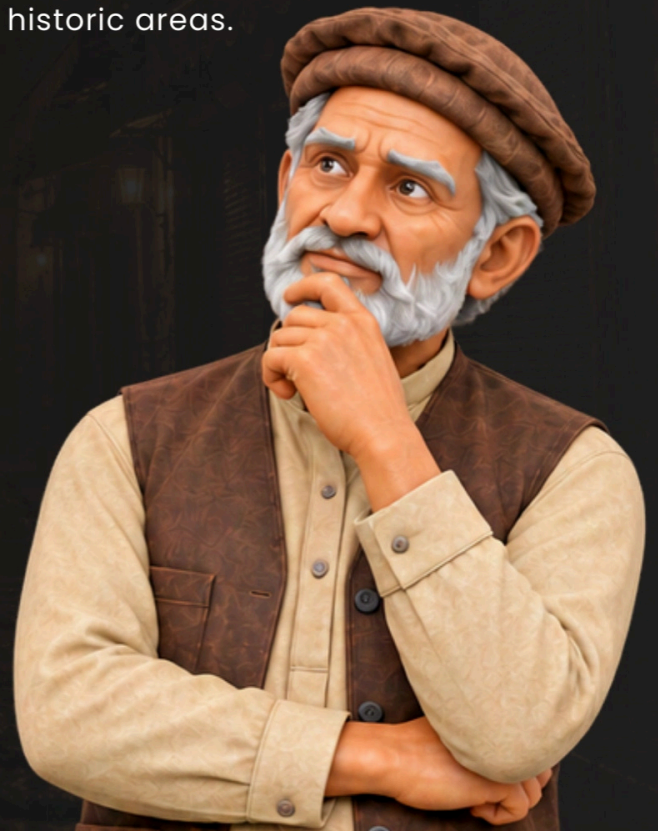
Primary research helped me understand how people currently experience Rawalpindi and what they would expect from an interactive cultural project.

SECONDARY RESEARCH

- The history of Rawalpindi
- Historic buildings and religious architecture
- Old neighbourhoods and markets
- Local food locations
- Rickshaw culture
- Pakistani textile and decorative art
- Environmental storytelling
- Interaction design in VR

SURVEY

I conducted a survey with 28 participants between the ages of 15 and 46.
The survey included people with different levels of familiarity with Rawalpindi. Some participants had lived in the city for many years, while others had limited knowledge of its historic areas.



CONCEPT DEVELOPMENT

INITIAL CONCEPT: A 3D ANIMATED FILM

The first version was a linear animation. It established the visual world, architecture, atmosphere and rickshaw journey.
Its main strength was cinematic storytelling. However, it provided no user choice.

FINAL CONCEPT: AN INTERACTIVE 3D AND VR JOURNEY GAME

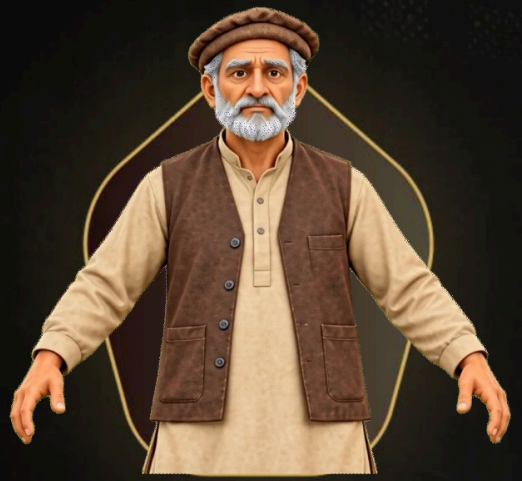
The final VR version places the user inside the rickshaw.

THE MAIN EXPERIENCE

AN INTERACTIVE 3D AND VR JOURNEY

1. The user begins the journey by selecting a driver and passenger.
2. After the characters are selected, the user enters the rickshaw from the passenger's perspective. The driver asks where they would like to go, and the user selects a route.
3. As the journey begins, the driver responds to nearby locations. The user can continue listening, stop the rickshaw, ask for more information or save a location in a digital memory journal

MEET THE DRIVERS



BABA JAMEEL

The Local Observer

- Friendly, practical and experienced
- Knows local streets and neighbourhoods
- Shares everyday stories about residents
- Best for the Market Route

RASHID

The History Keeper

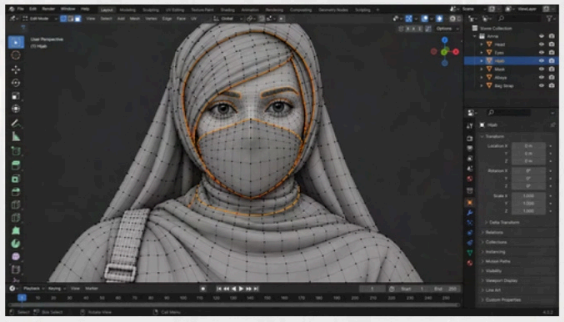
- Wise, calm and knowledgeable
- Knows old buildings and historical places
- Shares memories from Rawalpindi's past
- Best for the Architecture Route

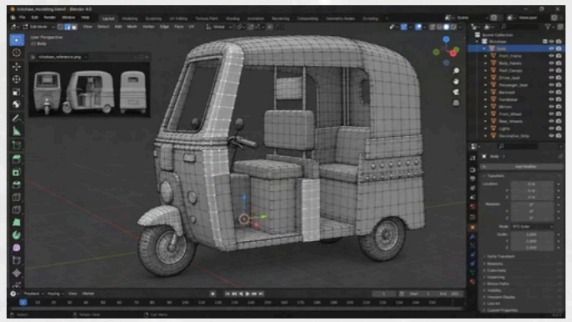
SAAD

The Local Observer

- Young, energetic and humorous
- Knows new food spots and city shortcuts
- Connects old Pindi with modern city life
- Best for the Food & Streets Route

ONE DRIVER ONE STORY





Conceptualization
Research
VISUALIZATION
IDEATION
Documentation

MEET THE PASSENGERS

HAMZA
The Young Professional

KHALA NASREEN
The Memory Keeper

AMNA
The Curious Student

SARA
The Cultural Explorer

UNCLE TARIQ
The Jolly Traveller

Passengers allow players to experience Rawalpindi through different personalities and perspectives. During the ride, they can select routes, look around, stop at important locations and ask the driver questions. Their choices determine which buildings, markets, food places and stories they discover.

ROUTE SYSTEM

The user is not forced to choose the recommended guide. Any driver can travel on any route, but the style of information changes according to the selected character.

ROUTE 1
ARCHITECTURE OF PINDI

This route focuses on:

- Historic façades
- Churches and religious buildings
- Havelis
- Wooden balconies
- Windows and decorative details
- Building materials
- Preservation and decay

Best guide: Baba Jameel

ROUTE 2
LIVING MEMORIES

This route explores:

- Family memories
- Childhood stories
- Neighbourhood traditions
- Markets and community life
- Sounds associated with particular streets
- Stories of how places have changed

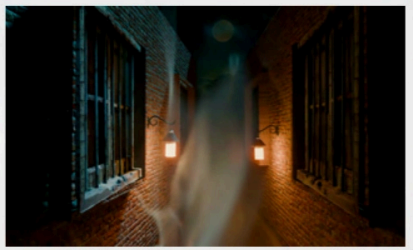
Best guide: Rashid

ROUTE 3
FOOD AND STREET

This route focuses on:

- Food stalls
- Tea shops
- Local restaurants
- Busy markets
- New gathering places
- Shortcuts and modern city movement

Best guide: Saad



MAMOO BURGER

MAMOO BURGER

INTERACTION DESIGN

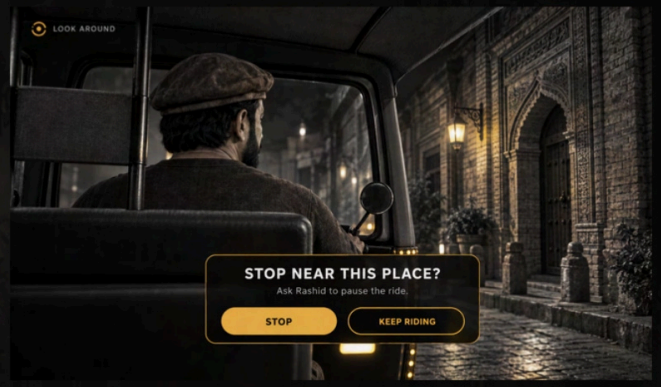
The interactions were kept simple so that users could focus on the environment.

ASK

A dialogue menu allows the user to request more information.

CONTINUE

The user can return to the ride whenever they are ready.

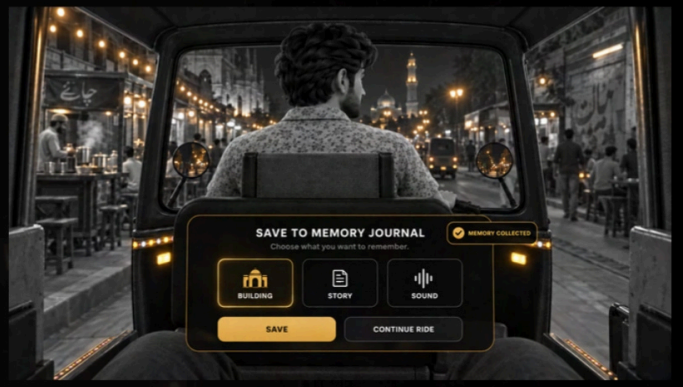


STOP

The user can ask the driver to stop near an important location.

LOOK

The user can look around the rickshaw and examine the street.

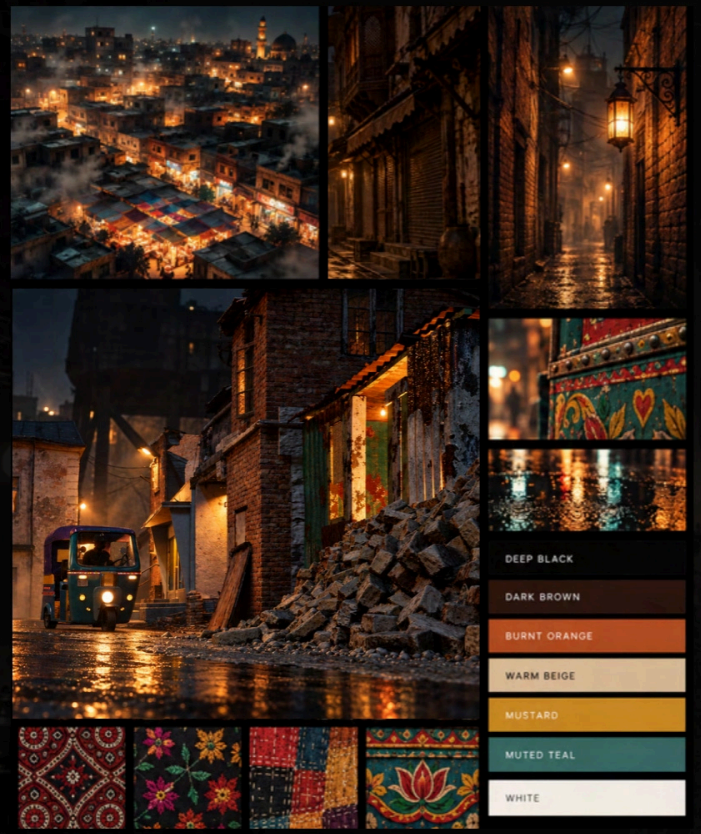


COLLECT

The user can save a building, story or sound in the memory journal.

VISUAL DIRECTION

- COLOUR PALETTE
- TEXTURES
- TYPOGRAPHY



GALIYAN PINDI DIYAN

VISUAL DIRECTION

AILERON

ABCDEFGHIJKLMNOPQRSTUVWXYZ

abcdefghijklmnopqurstuvwxyz

گلیاں پنڈی دیاں

CHARACTER ART DIRECTION

The characters are designed as **handcrafted 3D models**.
Their faces are expressive but not overly cartoonish or photorealistic. Slight imperfections in the modelling help them feel connected to the handmade visual language of the environment.
Clothing, body shapes, skin tones and accessories create variety while keeping the characters recognisably Pakistani.
The character designs avoid presenting everyone through the same age, body type or social identity.

ENVIRONMENTAL DESIGN

The environment is designed around observation rather than speed.
Important elements include:

- **Historic architecture** as the visual focus
- A moving rickshaw as the main viewpoint
- Warm bulbs against a dark night
- Winter fog and cooking smoke
- Reflections on wet streets
- Shop interiors and signs
- Layered environmental sound
- Architectural highlights triggered by proximity
- Street vendours

The environment should feel lived in, not empty or staged.

---------- EVERYDAY CHAI CULTURE OF ANDROON PINDI

INTERFACE DESIGN

The interface uses **minimal controls** and large **readable elements**.

BEGIN JOURNEY

BEGIN JOURNEY

BEGIN JOURNEY

SAVE TO MEMORY JOURNAL

SAVE CONTINUE

STOP NEAR THIS PLACE?

Ask Rashid to pause the ride.

STOP KEEP RIDING

RASHID

The Storyteller

- Local Memories
- Hidden Stories

MAIN INTERFACE COMPONENTS

Begin Journey button

Driver cards

Passenger cards

Route cards

Stop Here action

Ask Driver dialogue

Continue Ride action

Save Memory button

Sound and subtitle settings

Comfort settings

Memory journal

BABA JAMEEL ----------

Where would you like to go?

ARCHITECTURE OF PINDI

MARKET

FOOD & STREETS

DEVELOPMENT PROCESS

The existing 3D animation was used as the visual foundation for the VR game. Its environments, buildings, rickshaw, characters, lighting and cultural details were adapted into an interactive experience.

STEP 1

SCENE PREPARATION

The original Blender environment was cleaned and divided into explorable locations.

STEP 2

ASSET OPTIMIZATION

Buildings, characters and the rickshaw were optimized for smooth VR performance.

STEP 3

GAME ENGINE

The 3D assets were transferred to Unity

STEP 4

VR PERSPECTIVE

A 360° passenger viewpoint was placed inside the rickshaw.

STEP 5

GAME INTERACTIONS

Route selection, stopping, looking around, asking questions and collecting memories were added.

STEP 6

STORY SYSTEM

Driver dialogues and local stories were connected to important locations.

STEP 7

SOUND DESIGN

Rickshaw sounds, street noise, conversations and cultural music created an immersive atmosphere.

STEP 8

TESTING

Navigation, readability, comfort and performance were tested to improve the experience.

GAME FLOW

GALIYAN PINDI DIYAN
AN INTERACTIVE VR EXPLORATION GAME

CHOOSE YOUR DRIVER
EVERY DRIVER TELLS THE CITY DIFFERENTLY!
RASHID
STREETWISE LOCAL
Knows shortcuts, daily routes and local life.
BABA JAMEEL
HERITAGE GUIDE
Shares the history, buildings and forgotten stories of old Pindi.
SAAD
MODERN EXPLORER
Knows new landmarks, food spots and the changing city.
BACK
SELECT
NEXT

CHOOSE THE PASSENGER
CHOOSE HOW YOU WANT TO EXPLORE THE CITY!
HAMZA
The Young Professional
KHALA NASREEN
The Memory Keeper
AMNA
The Curious Student
SARA
The Cultural Explorer
UNCLE TARIQ
The Jolly Traveller
BACK
SELECT
NEXT

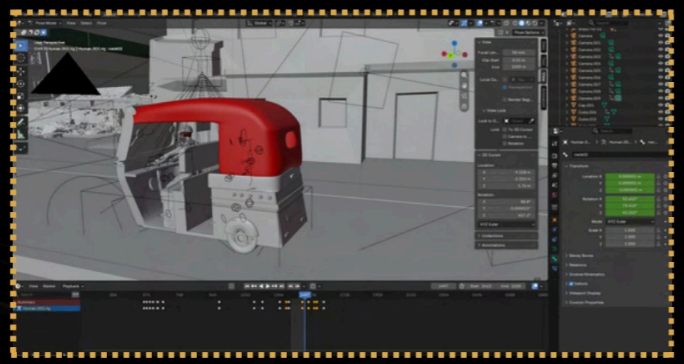
BABA JAMEEL
Where would you like to go?
ARCHITECTURE OF PINDI
MARKET
FOOD & STREETS
BACK
NEXT

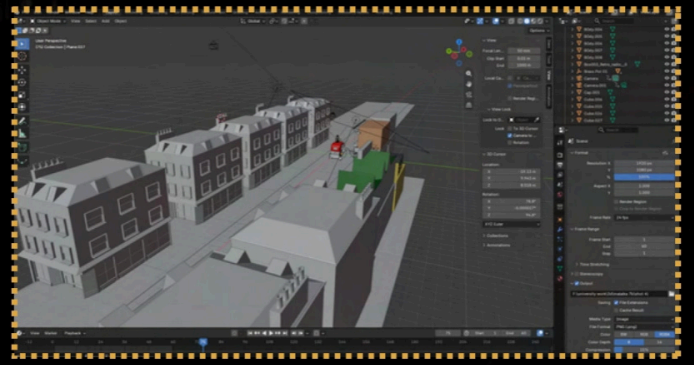
RASHID
These tight streets are my secret shortcuts through the heart of old Pindi.
KNOW MORE ABOUT IT?

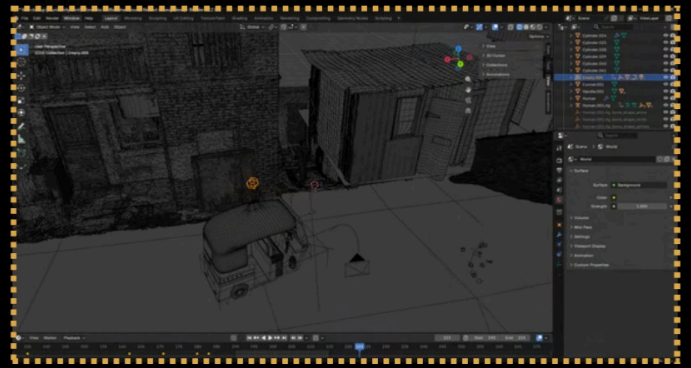
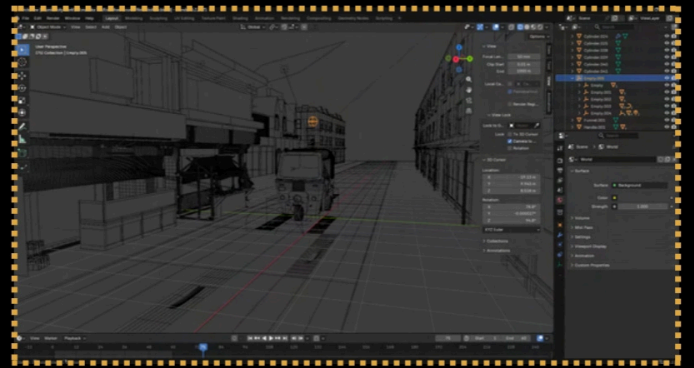
CHAI DHABA
LOCATION REACHED
MAMU BURGER
You are passing by Mamu burger, one of the city's most famous food spot.
Know more About it!

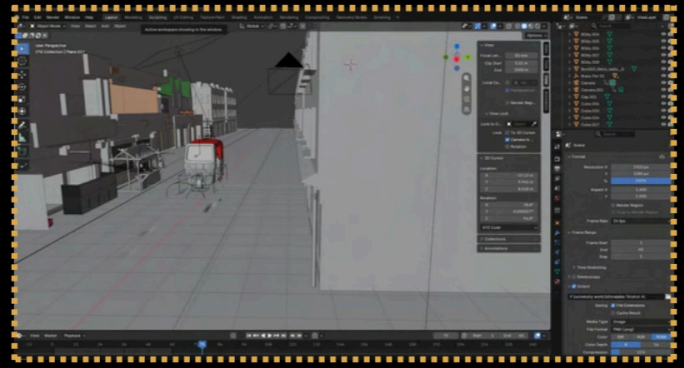
ENVIRONMENT DESIGN

I modelled the architecture and environment in **Blender** to recreate the character of Androon Rawalpindi. I focused on narrow streets, aged brick walls, traditional buildings and warm lighting to create an immersive and culturally authentic atmosphere.







DETAILS

These environment screenshots highlight the small details I added, including worn walls, brick textures, windows, street lights, fog and local decorative elements. Each detail helps make Androon Rawalpindi feel more realistic, atmospheric and culturally familiar.

Watch Video 01 - VR experience (click to play)

THE RICKSHAW

A rickshaw is a familiar form of local transport in Rawalpindi and an important part of the city's everyday life and visual identity.

HOW I MADE IT ?

I modelled the rickshaw in Blender, focusing on its basic structure, interior, seats, wheels and decorative details. I used vibrant colours and patterns inspired by Pakistani truck and rickshaw art.

WHY I USED IT ?

I chose the rickshaw because it offers a personal and culturally familiar way to explore Androon Rawalpindi. In the VR game, it becomes more than transport, it guides the player through narrow streets, architecture, local stories and everyday city life.

Watch Video 02 - Rickshaw tour (click to play)